

KM-BBB LAB SETUP

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Revision History

1.0	Basic Version created By Kishore kumar Boddu

1. Prerequisites

1.1. Shell Scripting & GIT videos

Watch and practise Shell scripting and GIT videos from Kernel Masters YouTube channel.

Shell Scripting Playlist Videos:

<https://youtube.com/playlist?list=PLt8joVUqRfUelaLzKosVBiUjcesrXHqBI>

GIT Playlist Videos:

<https://youtube.com/playlist?list=PLt8joVUqRfUd-CmYwRQdoNMXXDob0ErYD>

1.2. Test KM-BBB target board with the help of video.

Refer the video in www.youtube.com/watch?v=RLneJhUjCn0&t=1186s and test Kernel Masters BBB Expansion Board.

1.3. Learn how to use minicom (serial program) in Ubuntu

Refer internet and practise how to configure serial port parameters and how to use minicom more effectively.

1.4. Study Manuals and Prepare documents

Write down the answers in your note book the below things which we discussed in the last class.

Download embedded Linux documents from the below GITHUB repository.

<https://github.com/kernelmasters/embedded-linux>

Study and do the following tasks?

SoC (AM3358)

1. Write down specifications of am3358 controller?
2. Draw AM3358 block diagram?
3. Draw Memory Mapping of AM3358 controller?
4. Draw AM3358 Interrupt Vector table?

Target Board (BBB)

5. Write down specifications of beagle bone black board?
6. Draw Beagle Bone Black diagram?

KM-BBB

7. Write down specifications of KM beagle bone black board?
8. Draw KM Beagle Bone Black diagram?

2. Setup Embedded Linux Lab Environment

2.1. Experiment 1: Flash prebuilt images on SD card

Debian 10.3 prebuilt images flashing on two Micro SD cards using “km-bbb-sdcard-prebuilt-debian10.sh” script file and also understand the script file.

Prebuilt image location:

<https://github.com/kernelmasters/debian10-km-bbb.git>

First SD card: Stable images (prebuilt images)

Second SD card: Development images (own built images)

Third SD card: Yocto build images (own built images)

1. Create KM_GITHU folder and download debian repo.

```
$ mkdir ~/KM_GITHUB
$ cd ~/KM_GITHUB
$ git clone https://github.com/kernelmasters/debian10-km-bbb.git
$ cd ~/KM_GITHUB/debian10-km-bbb
```

2. Enter prebuilt Folder

```
$ cd ~/KM_GITHUB/debian10-km-bbb/prebuilt
```

3. Format SD Card with 1 partition and load boot loader images [MLO, u-boot.img] and debian 10 rootfs using the below script.

```
$ sudo ./km-bbb-sdcard-prebuilt-debian10.sh --mmc /dev/[drive]
```

"drive" is sdb or mmcblk0. find out using dmesg command after inserting sd card. script run 5 to 10 minutes.

4. Insert SDcard on BBB target and press switch2 and power on board. "km_bootmenu" choose sd card option and now BBB booting from SD card.
5. By default username:km; password:km enter login prompt. And update software packages otherwise sudo permission problem encounter.

```
$ sudo apt-get update
```

2.2. Experiment 2: Transfer SD Card Images in to on board memory eMMC

Copy SD card images in to on-board memory that is eMMC.

Script name & Location: /opt/scripts/tool/eMMC/

1. Insert SD card target board and boot from sdcard
2. go to scripts folder
\$ cd /opt/scripts/tool/eMMC/
3. Run script
\$ sudo ./init-eMMC-flasher-v3.sh

2.3. Experiment 3: Install ARM Cross toolchain

Install ARM Cross toolchain and supported packages to build Kernel source code. “*km-bbb-kernel-build.sh*”. After installation close terminal and start work on new terminal.

```
$ which arm-linux-gnueabi-gcc
```

2.4. Experiment 4: Test Hello World Application on target board

Write hello world application program and compile with ARM cross toolchain and transfer the binary file to target board using SD Card or scp command. Finally execute the binary file in KM-BBB expansion board and verify the results.

```
$ sudo cp hello-arm /media/km/rootfs/home/km/
$ sudo apt install net-tools ssh
```

3. Network Setup (How to connect BBB to the internet using Network cable)

3.1. Transfer files using scp

cp <source> <dest> (Within the system)

scp (secure Copy) (With in network)

Transfer from Laptop to KM_BBB:

```
scp <source> <userid>@<IP-Address>:~/
```

Transfer from KM_BBB to Laptop:

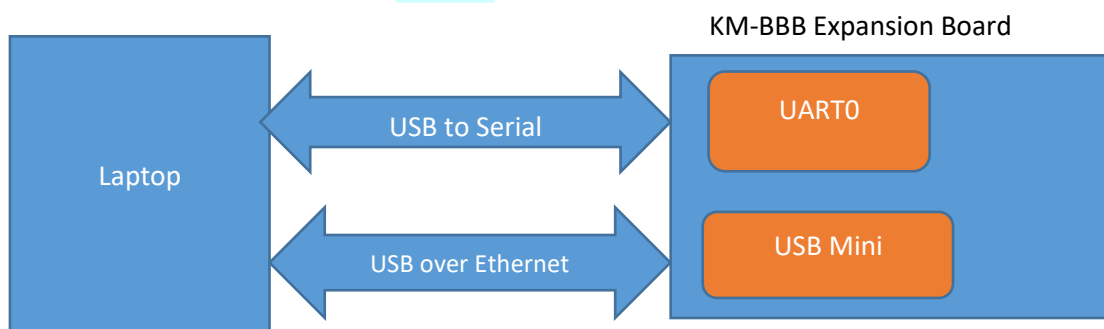
```
scp <userid>@<IP-Address>:~/smbus.c /home/km/Documents
```

```
scp hello-arm km@192.168.4.62:~/
```

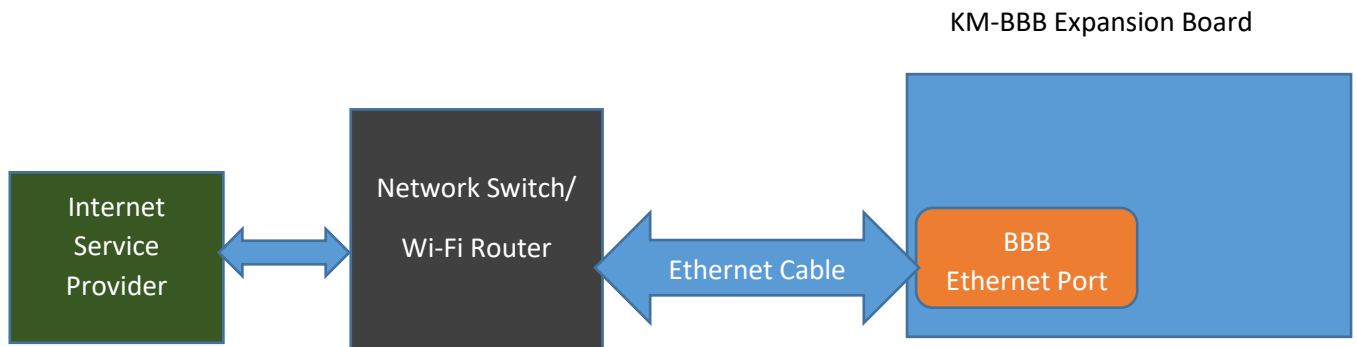
3.2. Remote login using ssh

From my laptop to BBB remote login

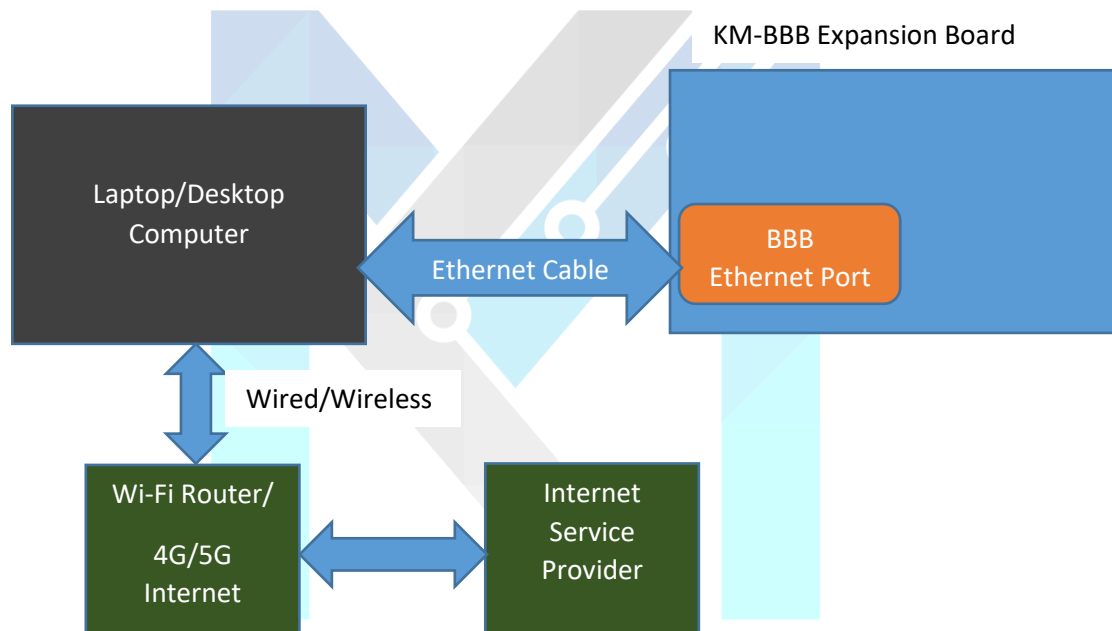
```
$ ssh <userid>@<IPAddress>
```



3.4. Connect BBB with Router or Network Switch using Ethernet Port (LAN)



3.5. Connect BBB with Laptop/Desktop using Ethernet Port



To enable Internet Connection Sharing in Windows 10, please follow the steps below:

1. Press **Windows key + X** to open the **Power User** menu and select **Network Connections**.
2. Right-click the network adapter with an Internet connection (Ethernet or wireless network adapter), then select **Properties**.
3. Click **Sharing**.
4. Put a check mark on **Allow other network users to connect through this computer's Internet connection**.
5. From the **Home networking connection** drop-down menu, select the **Microsoft Hosted Virtual Adapter**.
6. Click **OK** to finish.